MAT 221 - CALCULUS I
Lesson 13 - Numerical Integration

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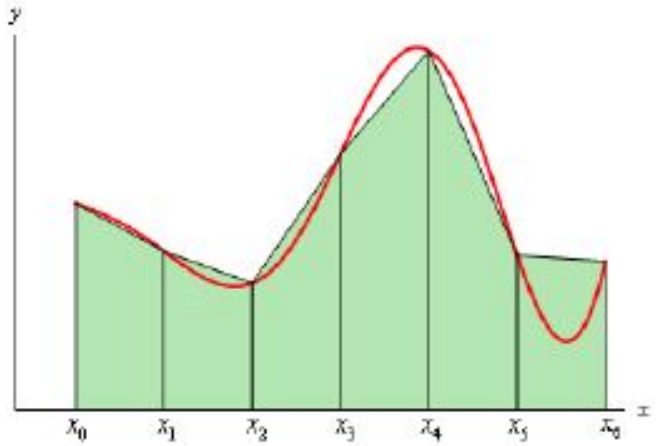
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Trapezoidal Rule

Let f be a function over an interval $[a, b]$ and partitioned by equally spaced points such that

$$a = x_0 < x_1 < \ldots < x_{i-1} < x_i \ldots x_n = b,$$

where $x_i - x_{i-1} = \Delta x = \frac{(b-a)}{n}$.
Then on each subinterval we will approximate the function with a straight line that is equal to the function values at either endpoint of the interval. see the figure below with $n = 6$



Each of these objects is a trapezoid, hence the rule's name. The area of the trapezoid in the $[x_{i-1}, x_i]$ is given by,

$$A_i = \frac{\Delta x}{2}(f(x_{i-1}) + f(x_i)).$$

So if we use n subintervals the integral is approximately,

$$\int_a^b f(x)\,dx \approx \text{sum of area of trapezoids}$$

$$= \frac{1}{2}[f(x_0) + f(x_1)]\Delta x + \frac{1}{2}[f(x_1) + f(x_2)]\Delta x + \ldots$$

$$+ \frac{1}{2}[f(x_{n-2}) + f(x_{n-1})]\Delta x + \frac{1}{2}[f(x_{n-1}) + f(x_n)]\Delta x. \quad (1)$$

The approximation formula (1) is called the trapezoidal rule for numerical integration. Since we have $\Delta x = \frac{(b-a)}{n}$, then we can write the trapezoidal rule as

$$\int_a^b f(x)dx \approx \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + \ldots + 2f(x_{n-1}) + f(x_n)] \quad (2)$$

Error Bound for Trapezoidal Rule

Suppose f is continuous on $[a, b]$ and f' and f'' exist on $[a, b]$, then the error bound for the trapezoidal rule is given by

$$|\epsilon_n^T| \leqslant M\frac{(b-a)^3}{12n^2} \tag{3}$$

where $M = \max_{x\in[a,b]} |f''(x)|$

Percentage Error

$$\text{Percentage error} = \frac{|A_{\text{true}} - A_n^T|}{A_{\text{true}}} \times 100\%$$

where $A_{\text{true}} = \int_a^b f(x)\,dx$.

Example 1

Estimate

$$\int_1^2 \frac{1}{x}dx$$

by using the trapezoidal rule with

(a) $n = 5$, and $n = 10$.

(b) Find the percentage error of the method for each n above.

Solution

(a) $n = 5$, and $\Delta x = \frac{b-a}{n} = \frac{2-1}{5} = 0.2$. Then

x_i	$f(x_i)$
1	1
1.2	0.8333
1.4	0.7143
1.6	0.625
1.8	0.5556
2.0	0.5

Hence

$$\int_1^2 \frac{1}{x} dx \approx A_5^T$$

$$= \frac{0.2}{2}[1 + 2(0.8333) + 2(0.7143) + 2(0.625) + 2(0.5556) + 0.5]$$

$$= 0.6956$$

(b) $A_{\text{true}} = \int_1^2 \frac{1}{x}\, dx = \ln x|_1^2 = \ln 2 \approx 0.6931$

$$\% \text{ error} = \frac{|A_{\text{true}} - A_5^T|}{A_{\text{true}}} \times 100\% = \frac{|0.6931 - 0.6956|}{0.6931} \times 100\% = 0.0036\%$$

For $n = 10$ to be done in class.

Example 2

(a) Find the error bound on discretizations error incurred when estimating

$$\int_2^3 \frac{1}{1-x}$$

by using the trapezoidal rule with 10 sub-intervals.

(b) Find the number of intervals to use in order to have a maximum error of 1 using the trapezoidal rule to estimate

$$\int_0^2 e^{x^2}\, dx.$$

(c) Use trapezoidal rule to approximate the integral

$$\int_1^2 \ln(1+x)\, dx$$

with a maximum error of 0.001.

Solutions

(a) $f(x) = \frac{1}{1-x}, f'(x) = \frac{1}{(1-x)^2}$ and $f''(x) = \frac{-2}{(1-x)^3}$. Hence, M is at $x = 2$ in the interval $[2, 3]$, that is, $M = |\frac{-2}{1-2}| = 2$. The error bound is

$$|\epsilon_n^T| \leqslant (2)\frac{(3-2)^3}{12(10)^2} = \frac{1}{600}.$$

(b) n should be chosen large enough so that $|\epsilon_n^T| \leqslant 1$. We have $f(x) = e^{x^2}, f'(x) = 2xe^{x^2}$ and $f''(x) = (2 + 4x^2)e^{x^2}$. Therefore our M occurs at $x = 2$. That is, $M = f''(x = 2) = 18e^4 \approx 983$. Hence,

$$|\epsilon_n^T| \leqslant (983)\frac{(2-0)^3}{12(n)^2} \approx \frac{655}{n^2} \leqslant 1.$$

This gives $n \geqslant \sqrt{655}$. The smallest n that meets the requirement is $n = 26$.

(c) Exercise